

Retrieval

Module 8 — BM25, dense, hybrid, and RAG

FINISH WELL

AI.SPIRE



GROWTH.

Keep learning.



CONSISTENCY.

Keep showing up.



COMMUNICATION.

Keep in touch.



PROGRESS.

Keep moving forward.

Let's do this together. 

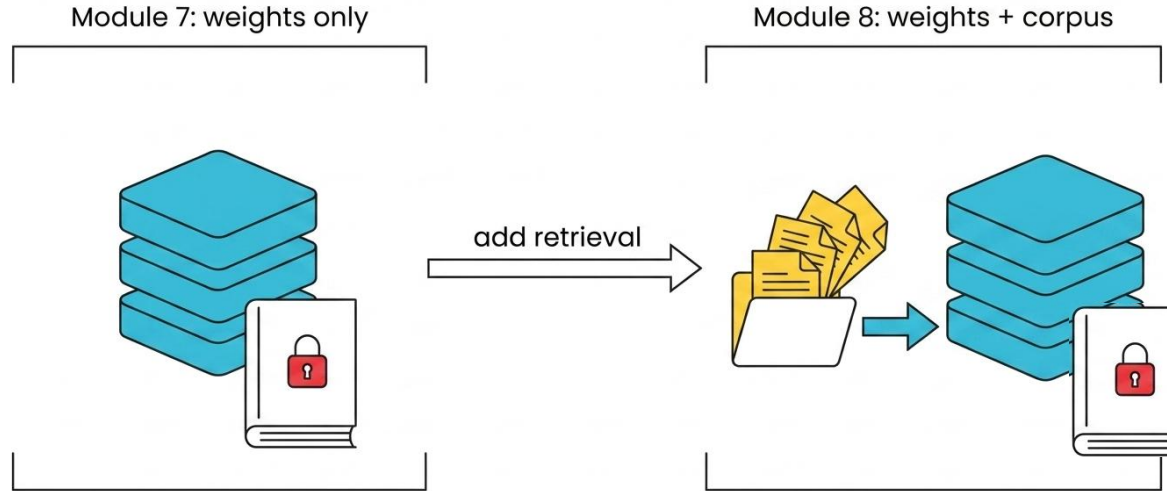
صندوق كفاءات المستقبل
Future Skills Fund



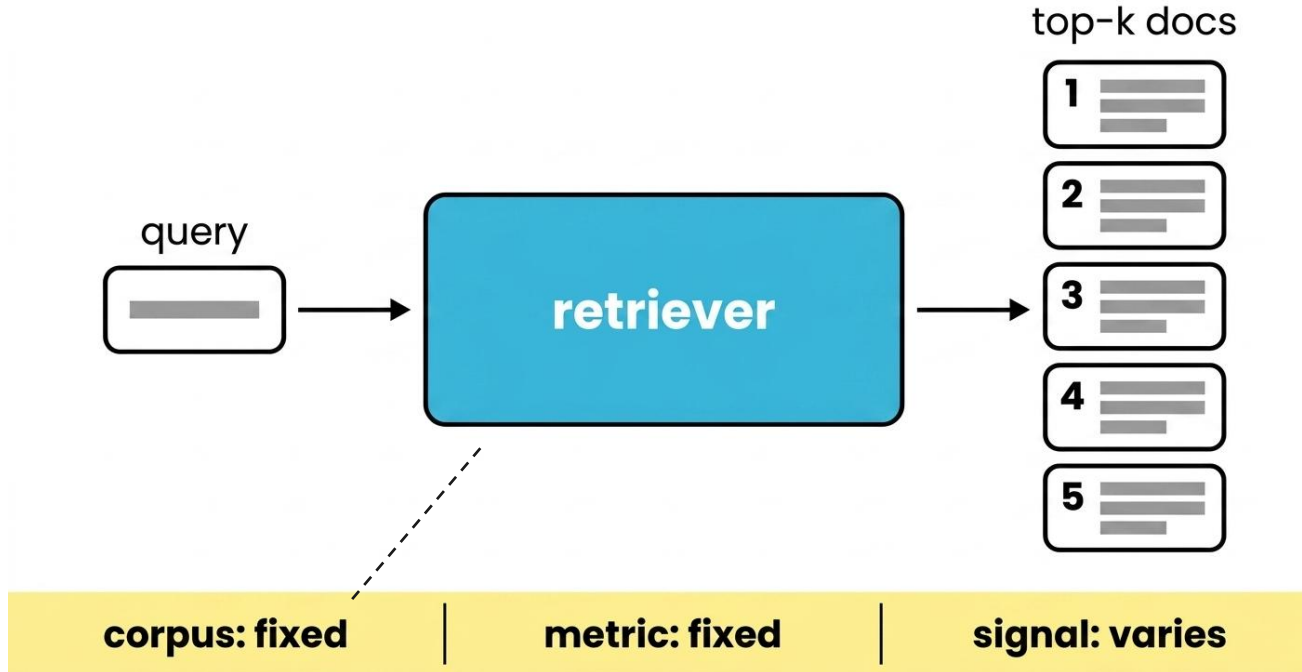
استدامة للاستشارات
Istidama consulting

LevelUp
Economy

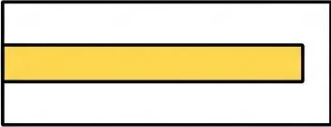
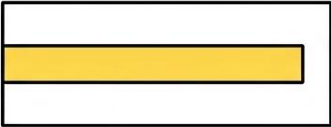
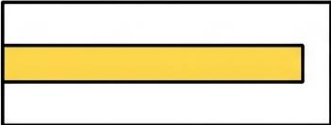
From Module 7 To Module 8



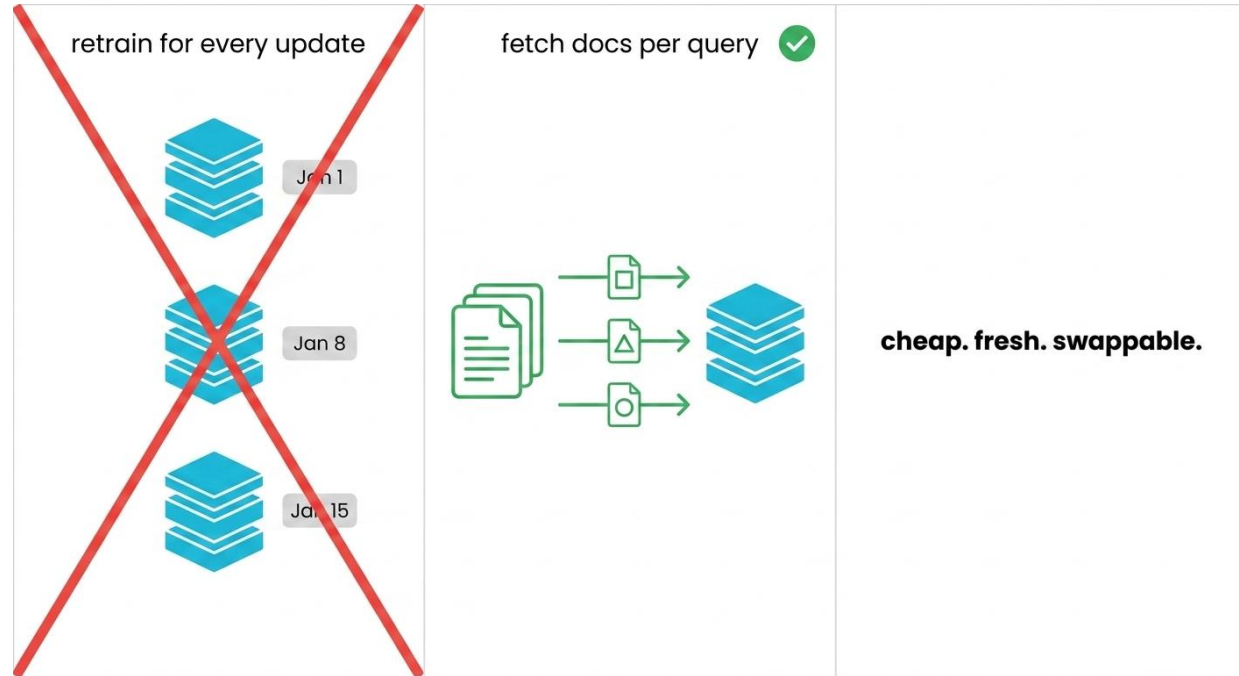
A Retriever Is A Function



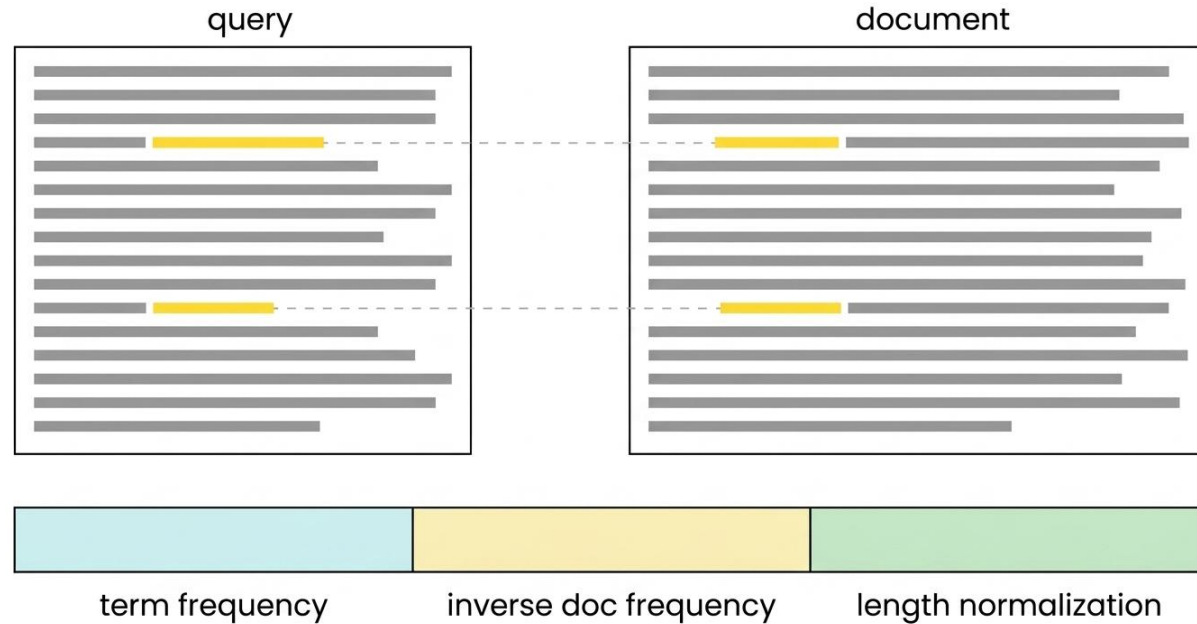
Three Queries, One Corpus

Query	BM25 sparse	Dense
rare token rare error code	 	 
paraphrase	 	 
rare token version string	 	 
	BM25: word match	Dense: meaning match

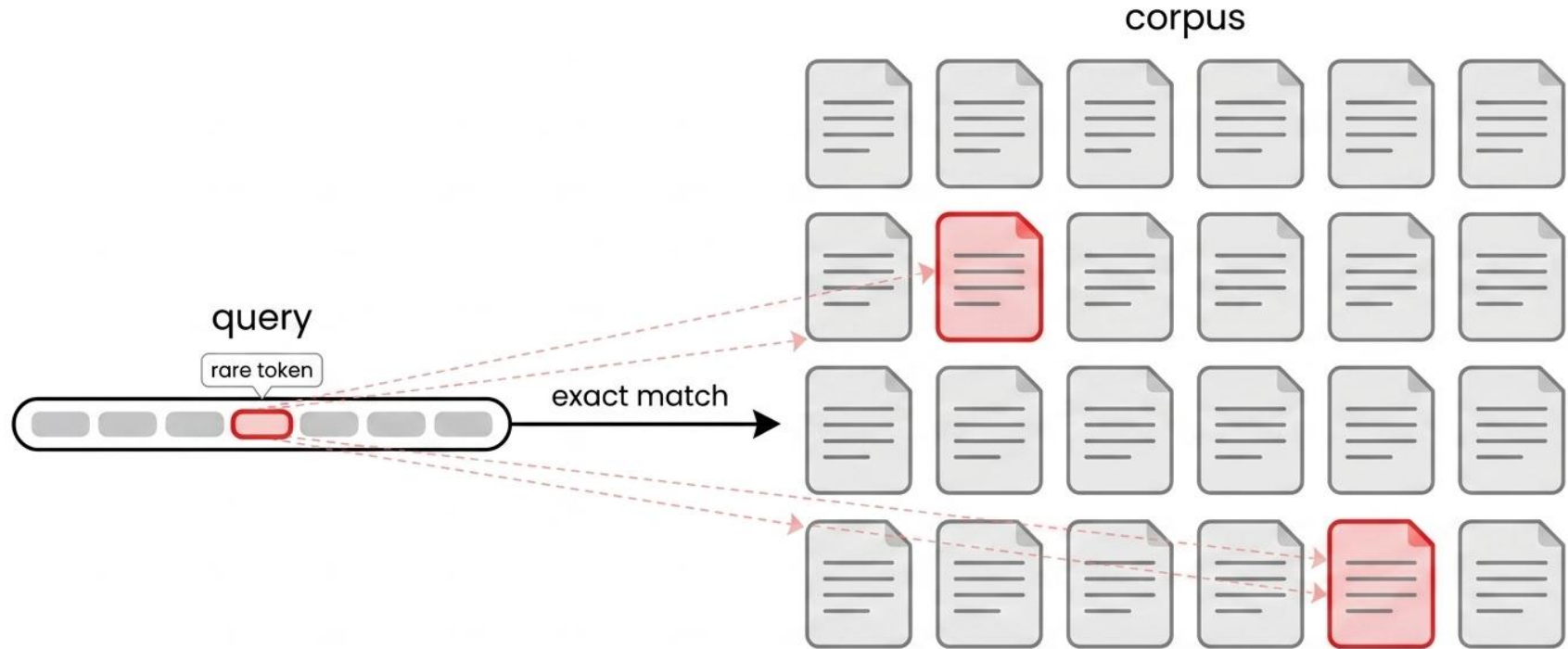
Why Use Retrieval



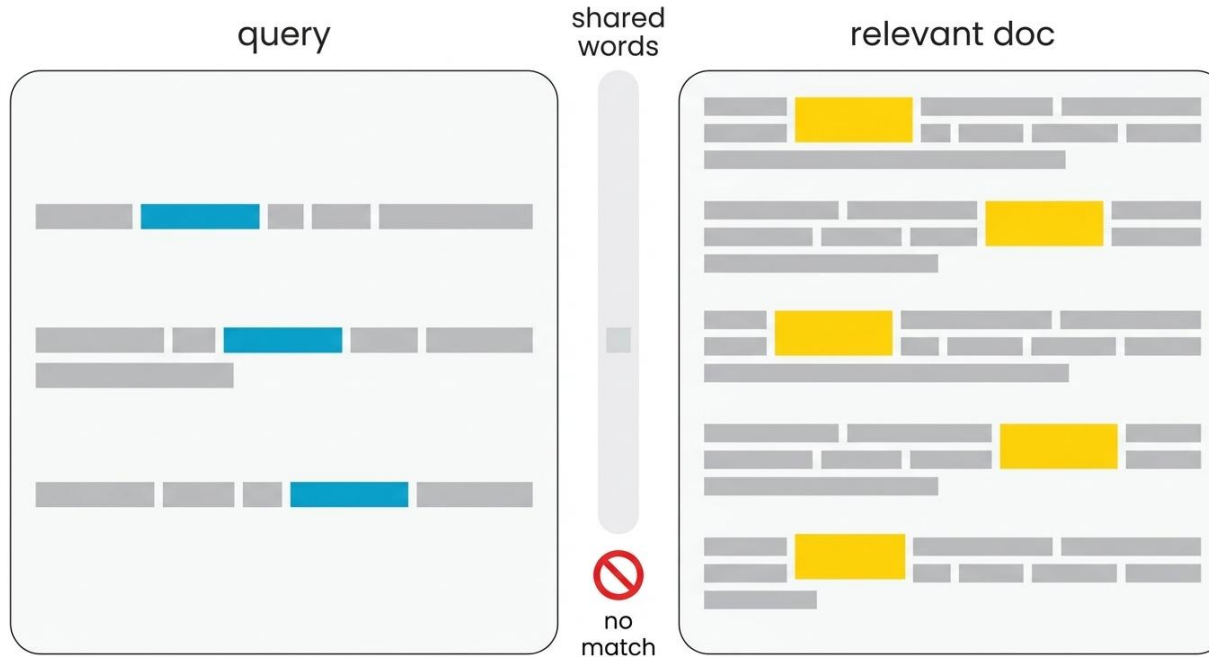
Sparse Signal: Words



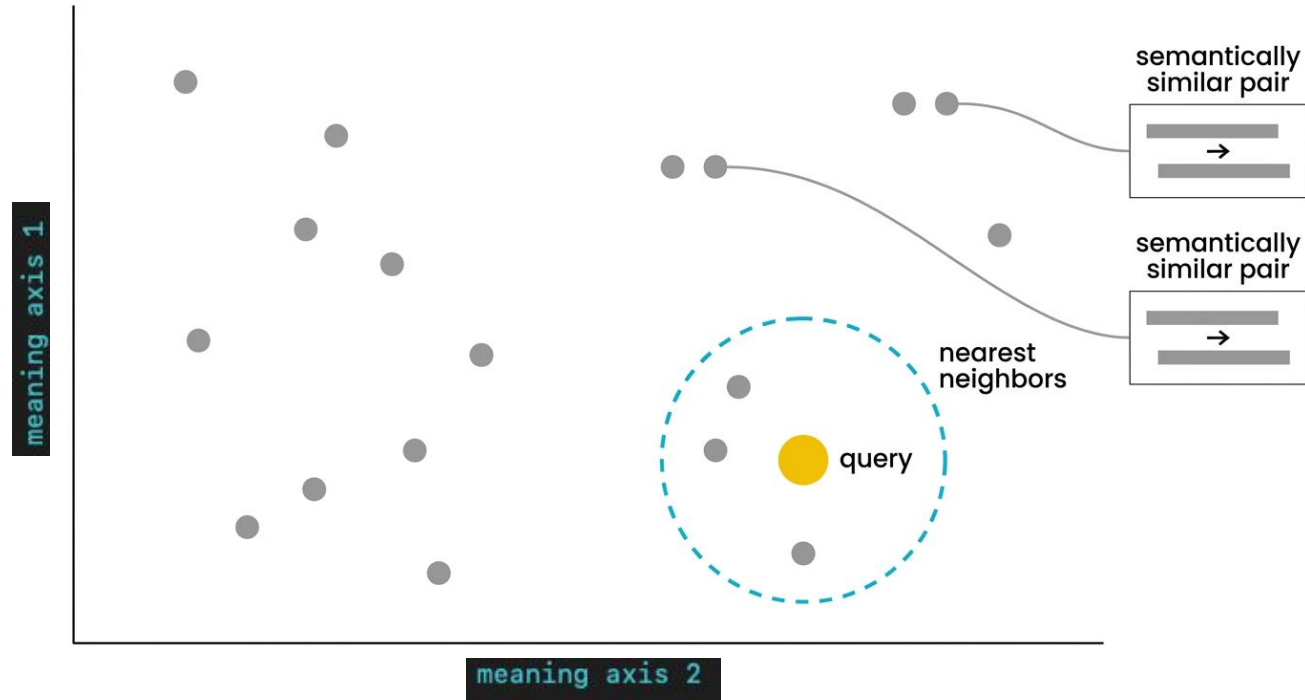
When BM25 is Great



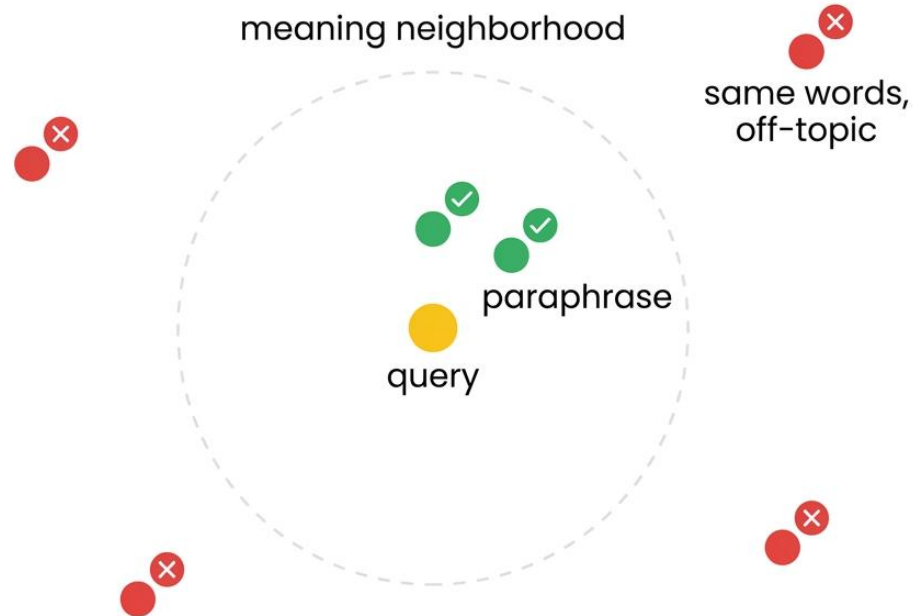
When BM25 Fails



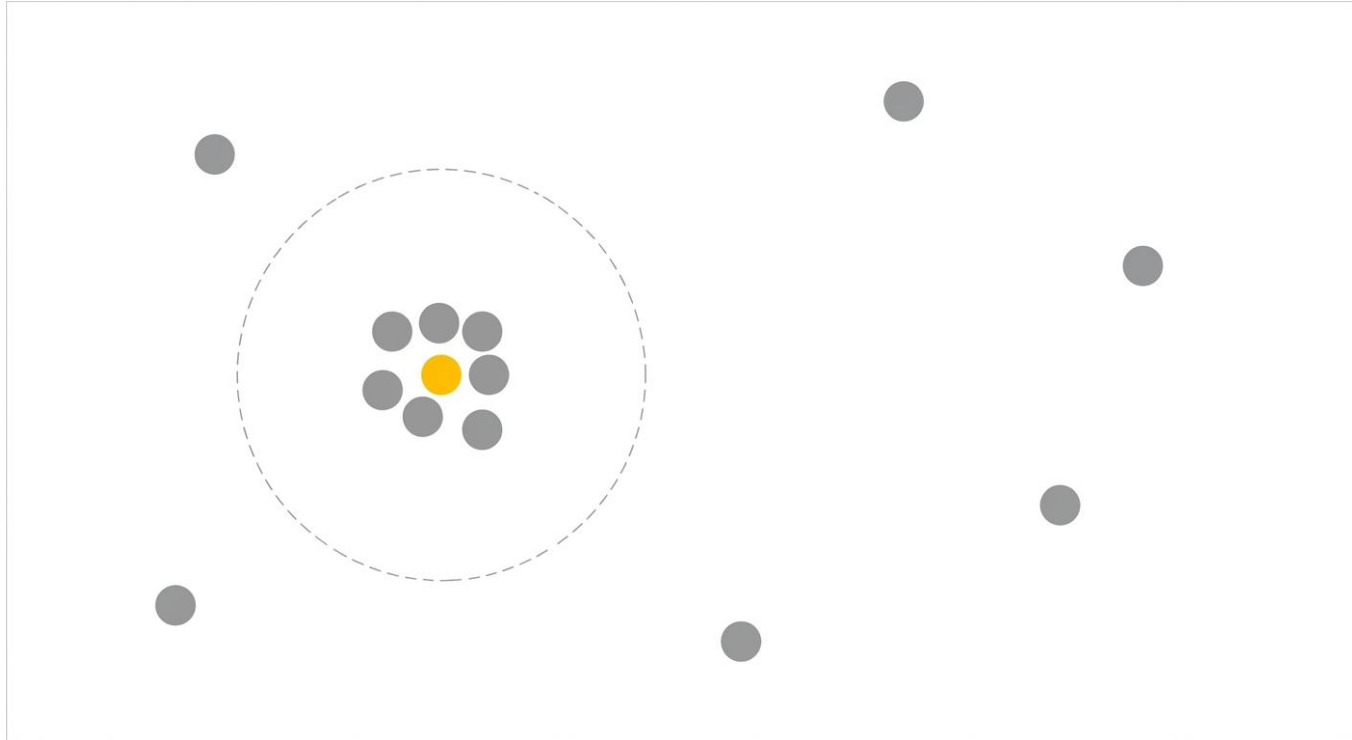
Dense Signal: Meaning



When Dense Helps



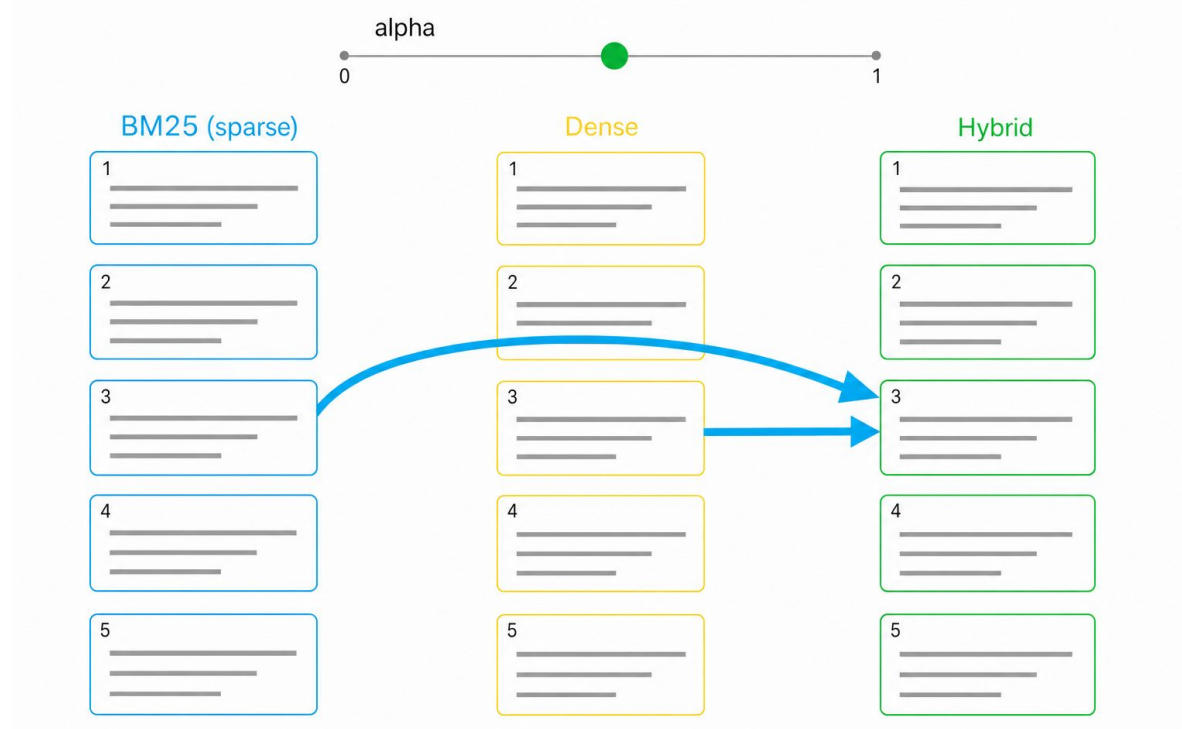
When Dense Fails



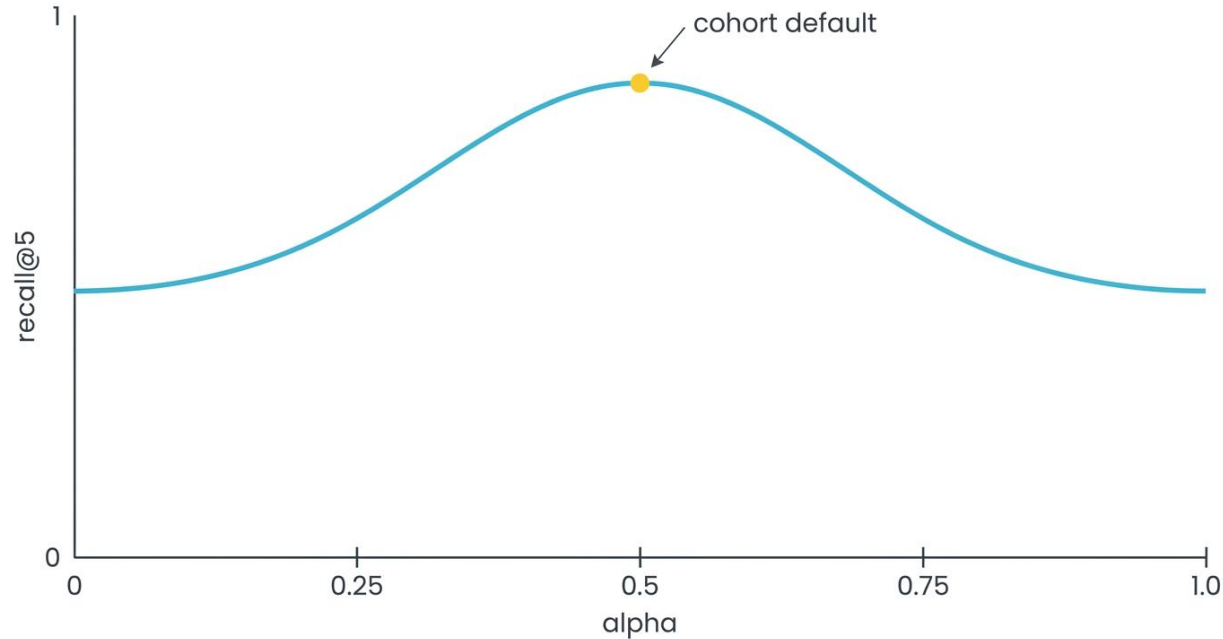
The Pattern

	BM25	Dense
rare-token query		
paraphrase query		
complementary, not redundant		

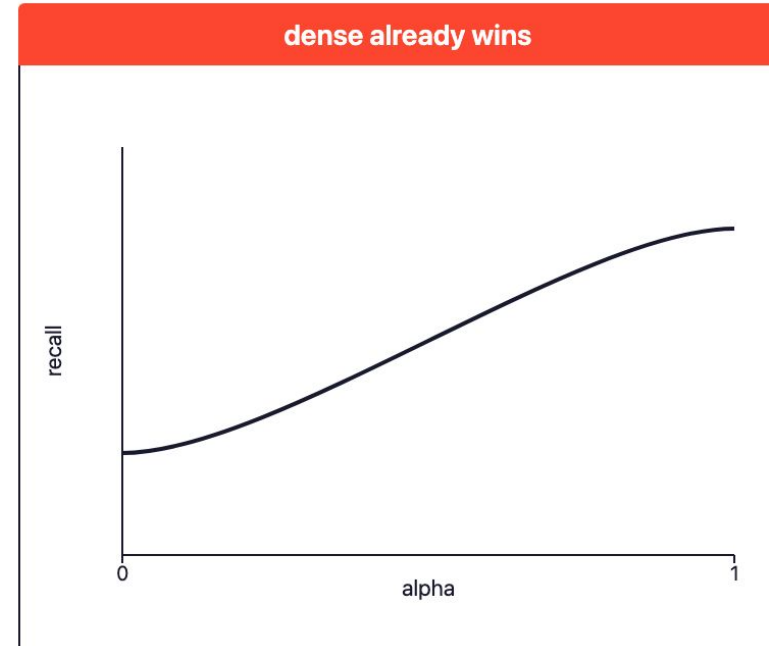
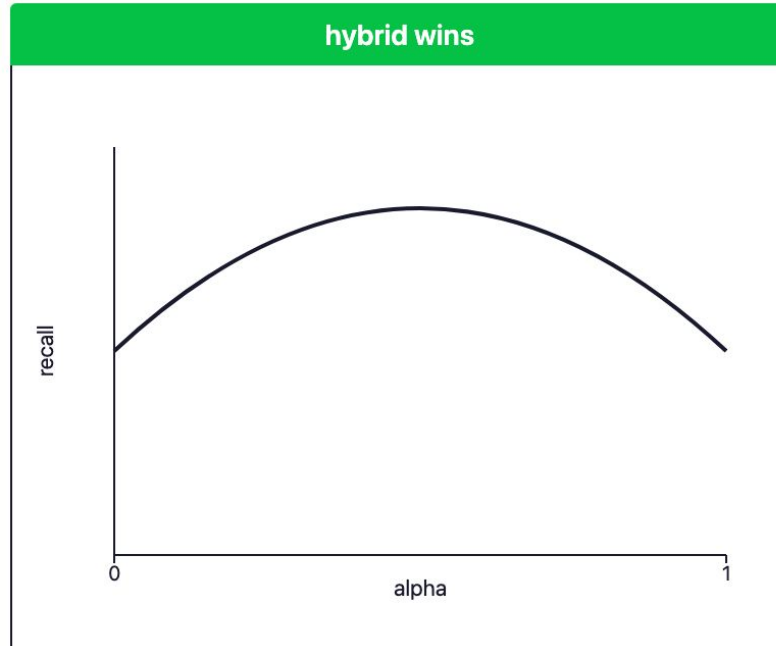
Hybrid: Run Both



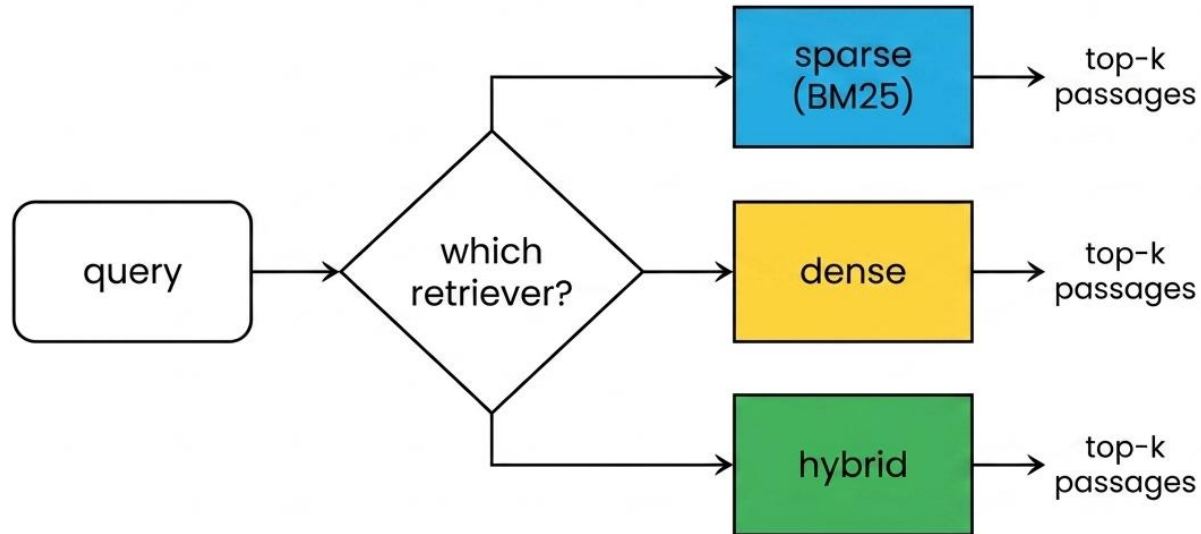
Choosing Alpha



Hybrid Is Not Free



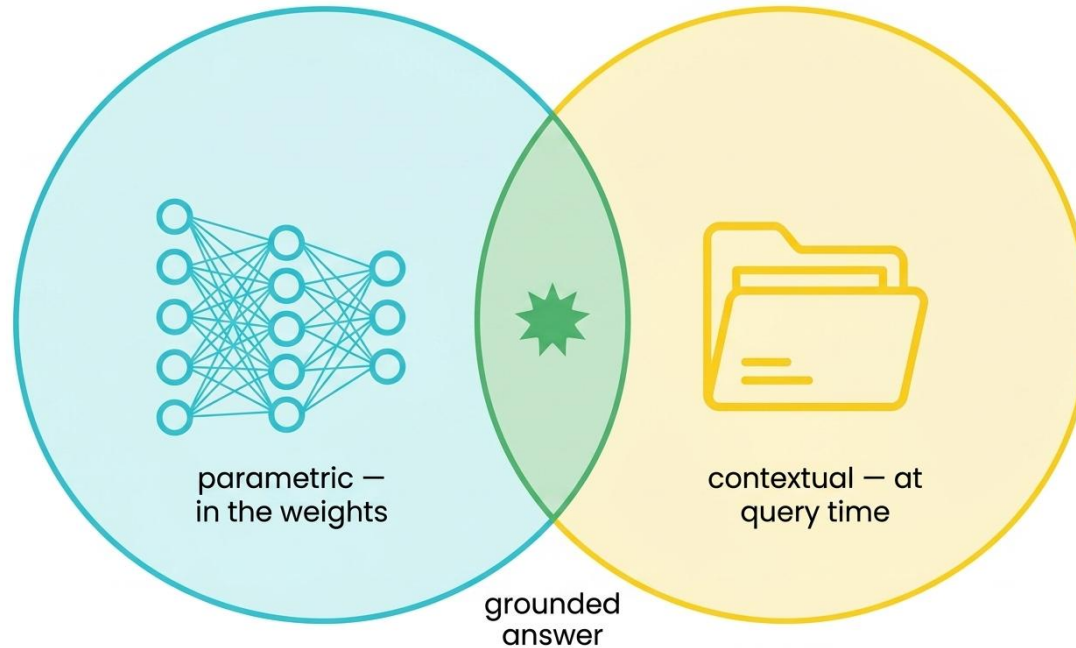
End Of Part One



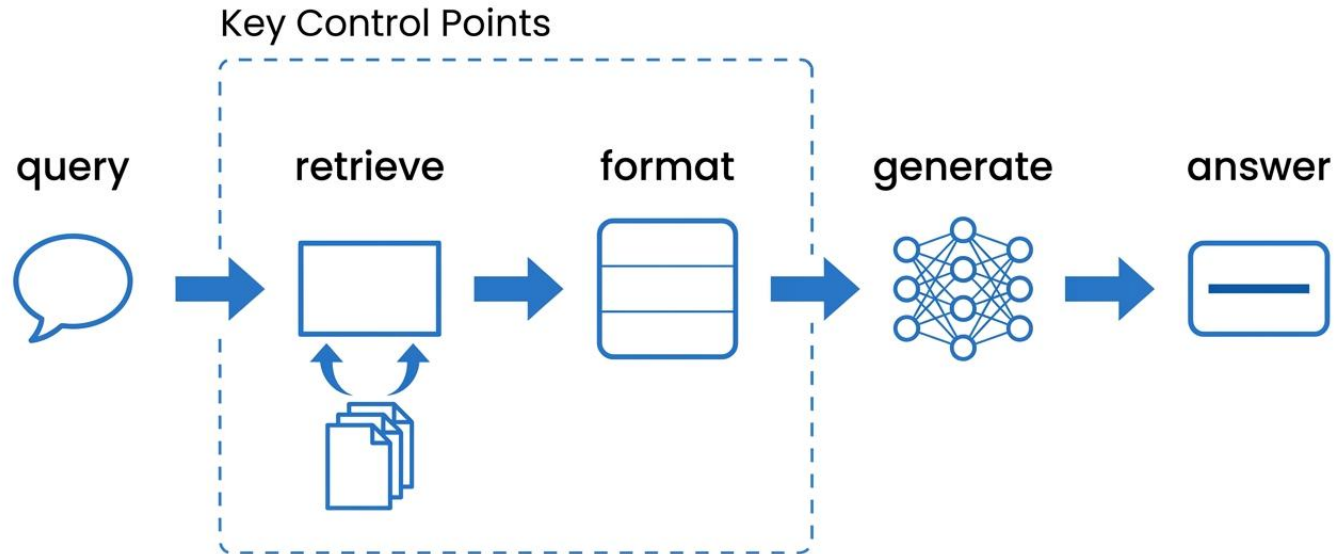
now what?

Parametric vs Contextual

`grounded answer = supported by retrieved context`



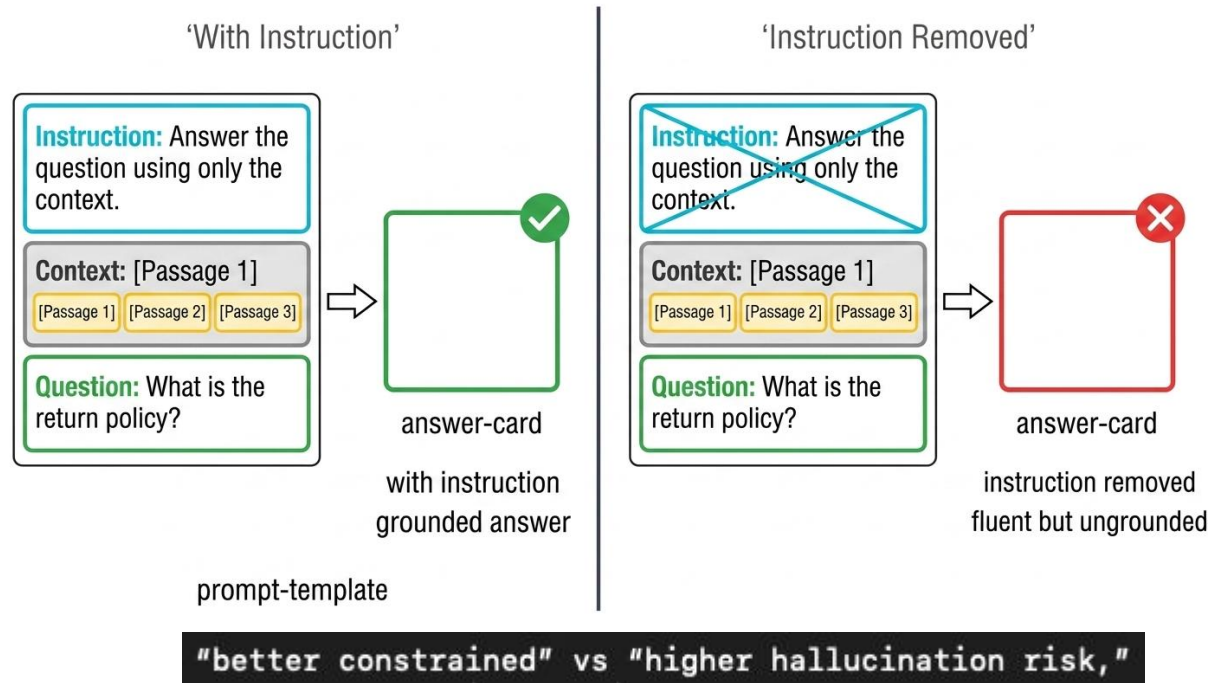
RAG In Five Steps



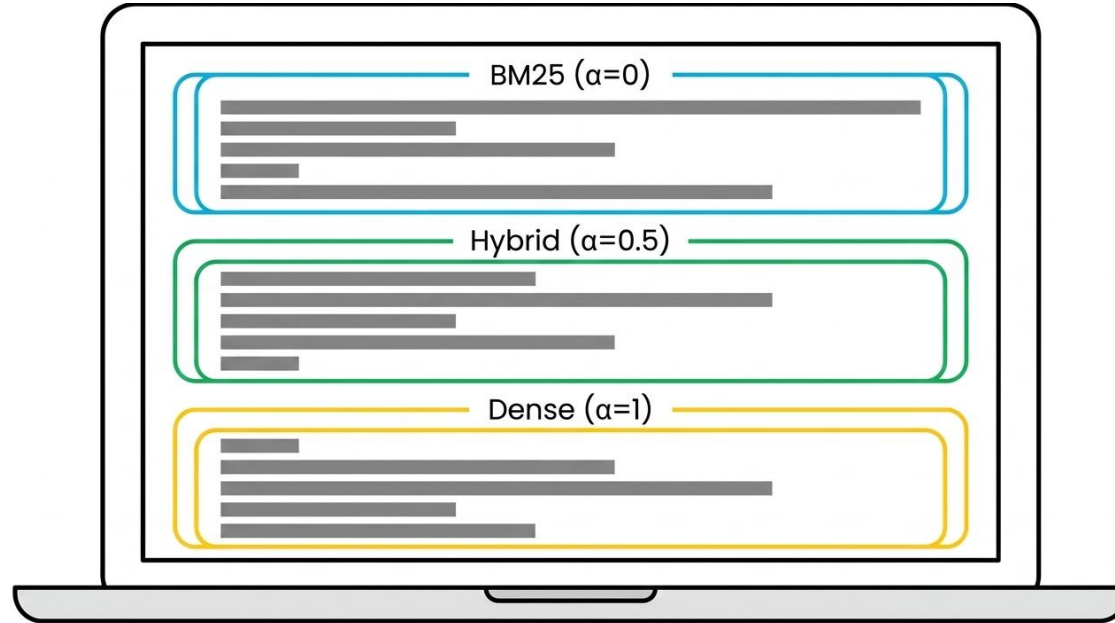
The Prompt Template

Instruction	Instruction _____
Retrieved Context	Context [1] _____ [2] _____ [3] _____ [4] _____ [5] _____
Question	Question _____ Answer:

The Instruction Earns Its Keep

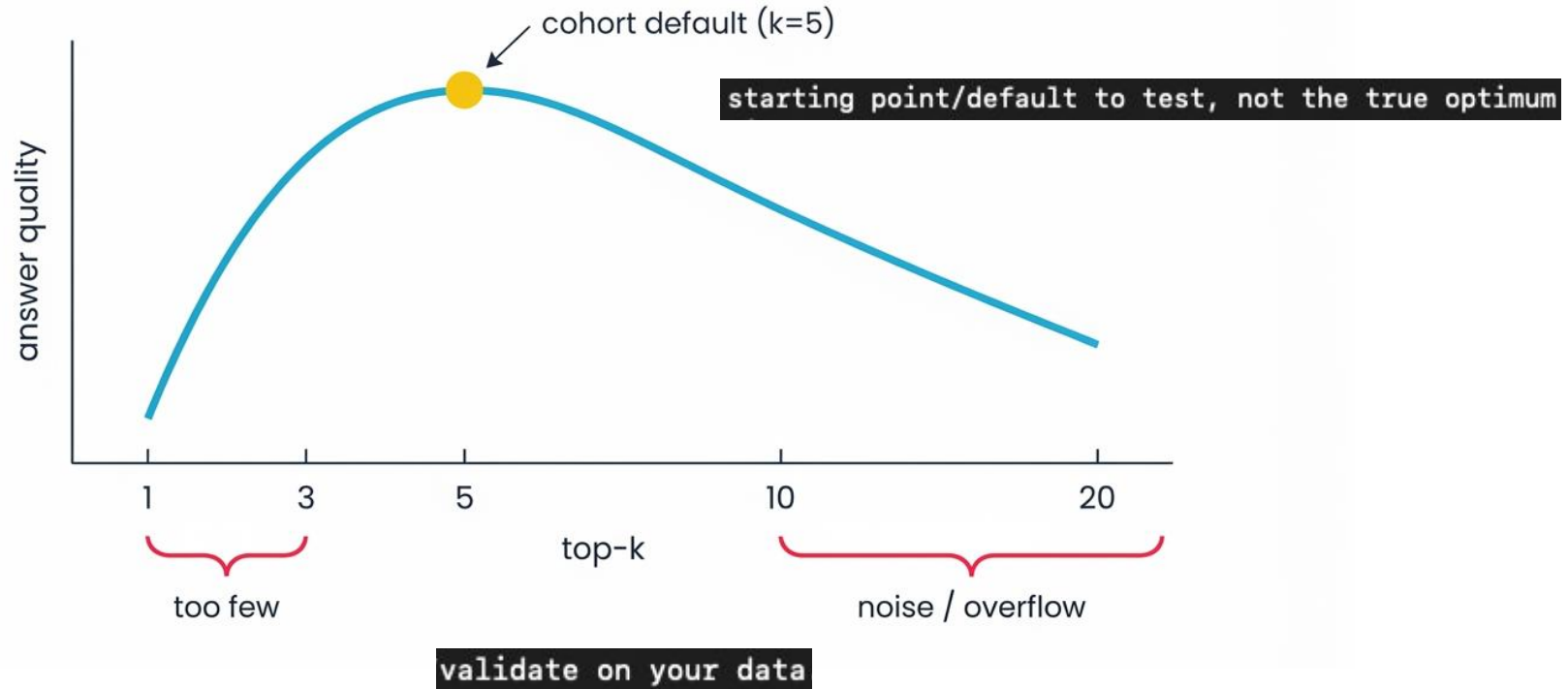


Optional: See It Run

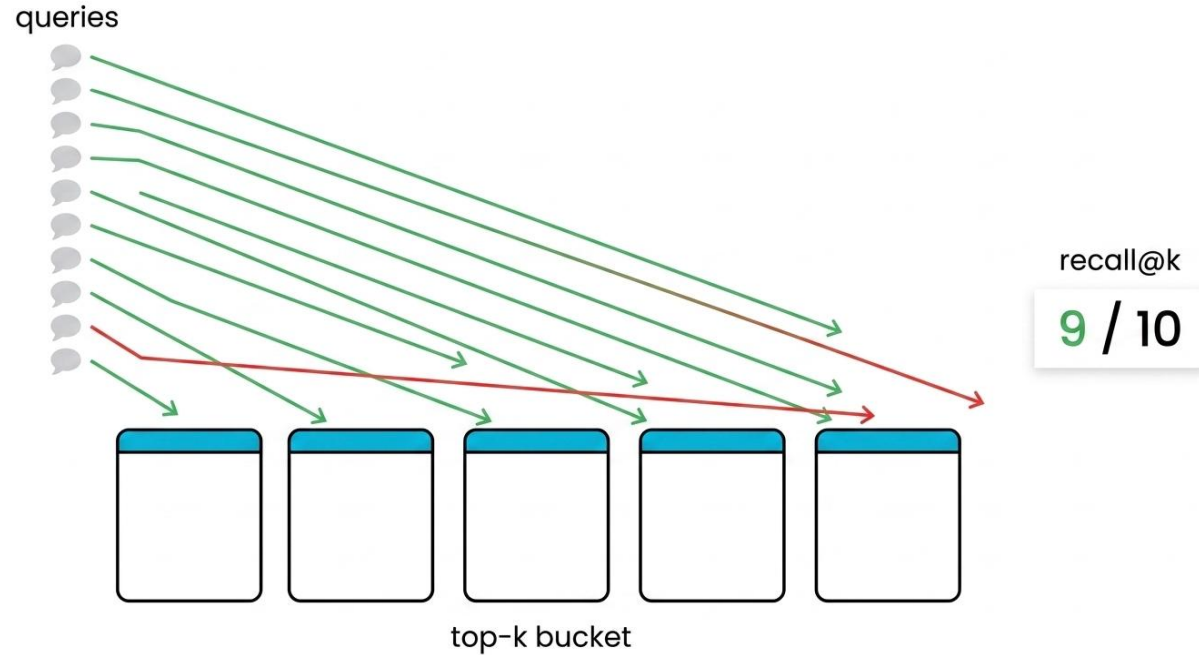


same corpus, three rankings

Choosing K

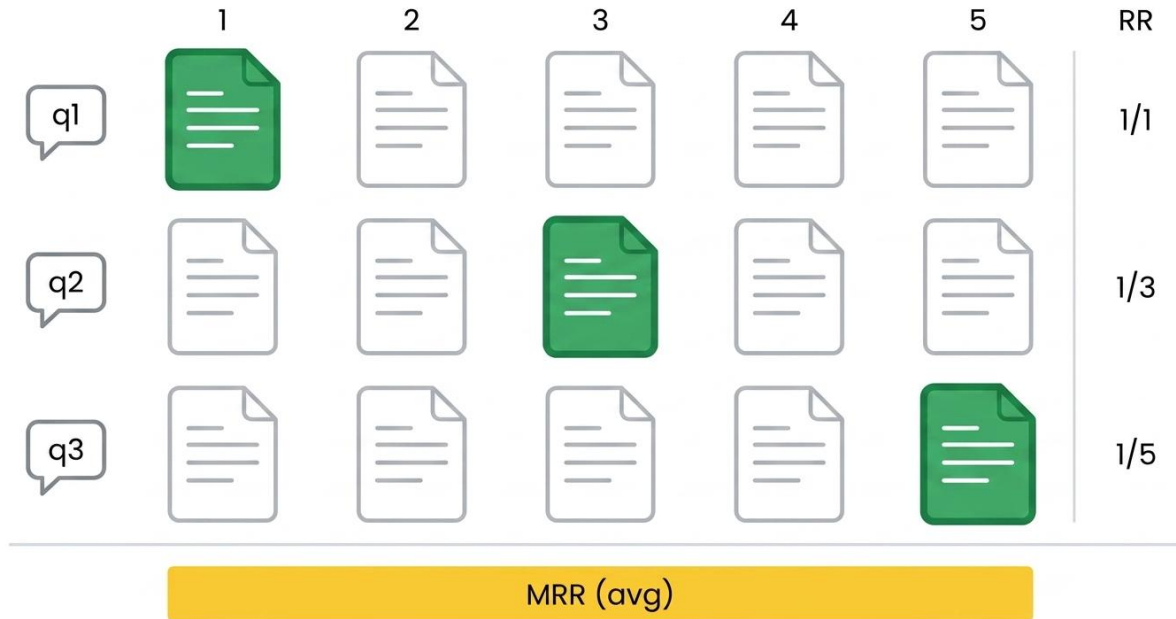


Recall At K

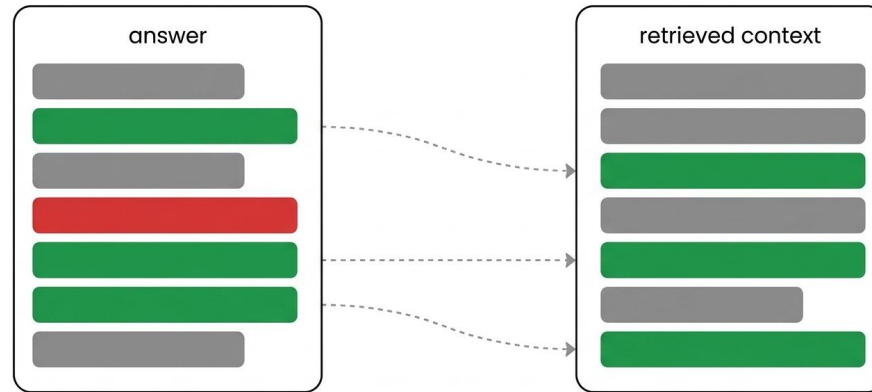


Mean Reciprocal Rank

$$\text{MRR} = (1 + 1/3 + 1/5) / 3 = 0.51$$



Groundedness



3/4 content words match → grounded

grounded (≥ 0.6)

watch-out (≤ 0.4)

"likely grounded," "grounding signal," or "review pass."

Where This Week Lands

	BM25	Dense	Hybrid
Query 1: Rare error code	1. doc-10A ✓ 2. doc-10B 3. doc-10C	1. doc-11X ✗ 2. doc-11Y 3. doc-11Z	1. doc-10A ✓ 2. doc-10B 3. doc-11X
Query 2: Paraphrase	1. doc-20L ✗ 2. doc-20M 3. doc-20N	1. doc-21P ✓ 2. doc-21Q 3. doc-21R	1. doc-21P ✓ 2. doc-20L 3. doc-21Q
Query 3: Version string	1. doc-30U ✓ 2. doc-30V 3. doc-30W	1. doc-31S ✗ 2. doc-31T 3. doc-31U	1. doc-30U ✓ 2. doc-31S 3. doc-30V



Questions?

Announcements

Congratulations!



This week's Module Mentors

- Marwan
- Nayef
- Shahd
- Hosam
- Amer
- Bashar